

Development and Validation of an Integrated Deep Learning Framework for 3D Carotid Artery Reconstruction and Stenosis Detection using MR angiography

Purpose: This study aimed to develop and validate an integrated deep learning framework for high-fidelity 3D vascular reconstruction and AI-based detection of carotid artery stenosis using Time-of-Flight (TOF) Magnetic Resonance Angiography (MRA).

Methods: The proposed framework consisted of two primary modules: (1) a volumetric segmentation network based on a 3D U-Net architecture, trained using 200 curated carotid MRA datasets for precise vascular reconstruction, and (2) a stenosis detection algorithm trained on an annotated dataset comprising 622 cases with varying degrees of stenosis and complete occlusion. Diagnostic performance was evaluated using 450 carotid MRA examinations (300 normal controls and 150 pathological cases), and AI-generated outputs were compared with expert reference standards.

Results: The reconstruction module demonstrated high anatomical fidelity, achieving a Dice similarity coefficient of 0.916 with a mean processing time of 4.5 seconds per case. For the detection of clinically significant carotid stenosis and occlusion, the framework achieved a sensitivity of 85.3% (128/150 cases) and a specificity of 92.3% (277/300 cases), indicating strong diagnostic accuracy and reliability.

Conclusion: The integrated AI framework significantly enhances diagnostic precision for neurovascular diseases by identifying subclinical or occult lesions, such as subtle stenotic segments, which are frequently susceptible to clinician fatigue or oversight. This solution serves as a potent clinical decision-support tool, optimizing diagnostic workflows and mitigating inter-observer variability in high-volume neurovascular practice.